

ENGINEERING CHANGE NOTE (ECN)

PROJECT: PROJECT-EEPROM-SEMA (based on PROJECT-USERTOS)

ORIGINATION: Kidd, Head Technical Lead, UMBC

PURPOSE OF CHANGE: Add EEPROM semaphore-managed data frame allocation

DOCUMENT NUMBER: ECN4.1

DATE OF PUBLICATION: April 30, 2026

LEAD ENGINEER:

Nathan Demory, UMBC

TEAM LEAD:

Nathan Demory

EXECUTIVE SUMMARY:

This ECN documents the changes required to add EEPROM-based data frame management to PROJECT-USERTOS. No customer requirements have changed. One new high-level requirement (HL.21) has been added requiring a mechanism for tasks to save sensitive data across power interruptions. Six mid-level requirements (ML.21.2–ML.21.6) further define the design: three 256-byte frames in EEPROM, managed via a FreeRTOS counting semaphore. One new educational constraint (E.4.5) requires demonstration of allocation and release by multiple tasks. New tests T.18–T.24 validate the EEPROM frame system.

HIGH-LEVEL DESCRIPTION:

Add a counting semaphore-managed EEPROM frame allocator to the existing FreeRTOS task system. Tasks acquire, use, and release 256-byte EEPROM frames to preserve data across power interruptions.

REASON FOR CHANGE:

ECR 5 (issued by Kidd, 4/20/2026) requires implementation of HL.21: a mechanism must exist whereby tasks can temporarily save sensitive data even across power interruptions. The design team extended this to mid-level requirements ML.21.2–ML.21.6 specifying EEPROM frames managed by a semaphore pattern.

CHANGED DOCUMENTS:

PROJECT REPORT (Demory.cmpe311.spring26.project#5):

- High-Level Requirements: Added HL.21 (4/30/2026, Demory — per Kidd ECR 5, 4/20/2026)
- Mid-Level Requirements: Added ML.21.2–ML.21.6 (4/30/2026, Demory)

- Educational Requirements: Added E.4.5 (4/30/2026, Demory — per Kidd, 4/20/2026)
- Testing Requirements: Added T.18–T.24 (4/30/2026, Demory)
- Definitions: Added EEPROM, Data Frame, Counting Semaphore, acquireFrame, releaseFrame (4/30/2026, Demory)
- Design Pre-requisites: Added EEPROM library note (4/30/2026, Demory)
- Design Overview: Added EEPROM frame layout and semaphore design sections (4/30/2026, Demory)
- Testing table: Added rows T18–T24 (4/30/2026, Demory)

ADDED DOCUMENTS:

project5_eeprom.ino — Updated Arduino sketch adding EEPROM frame management on top of FreeRTOS task architecture from project#4.

REMOVED DOCUMENTS:

None — project4_freertos.ino retained in repository for reference.

REPLACED DOCUMENTS:

Demory.cmpe311.spring26.project#4 replaced by Demory.cmpe311.spring26.project#5 incorporating HL.21, ML.21.2–ML.21.6, E.4.5, and T.18–T.24.

ANY ADDITIONAL DESIGN CONSIDERATIONS:

EEPROM wear:

- ATmega328p EEPROM is rated for ~100,000 write cycles per address.
- EEPROM.update() is used instead of EEPROM.write() to skip writes when value is unchanged, extending lifespan.
- LED tasks write on every blink cycle. At 300ms interval, this is ~115,200 writes/day per address. Consider writing only on interval change for production use.

Semaphore design:

- xFrameSemaphore: counting semaphore, max=3, initial=3
- xFrameMutex: binary mutex protecting frameOwner[] array
- acquireFrame() is blocking — a task will wait indefinitely if all 3 frames are held
- With 3 tasks (LED1, LED2, Fan) and 3 frames, all tasks can hold one frame simultaneously with none left over — this is intentional per ML.21.4 (at least 3 frames)

Memory note:

- Stack sizes increased to 192 words per task to accommodate EEPROM API call depth
- Total SRAM usage: FreeRTOS kernel ~200B + tasks (256+192+192+192=832 words = 1664B) — monitor for stack overflow

NOTE: All additions made on 4/30/2026 by Nathan Demory per ECR 5 issued by Kidd
4/20/2026.

END OF DOCUMENT